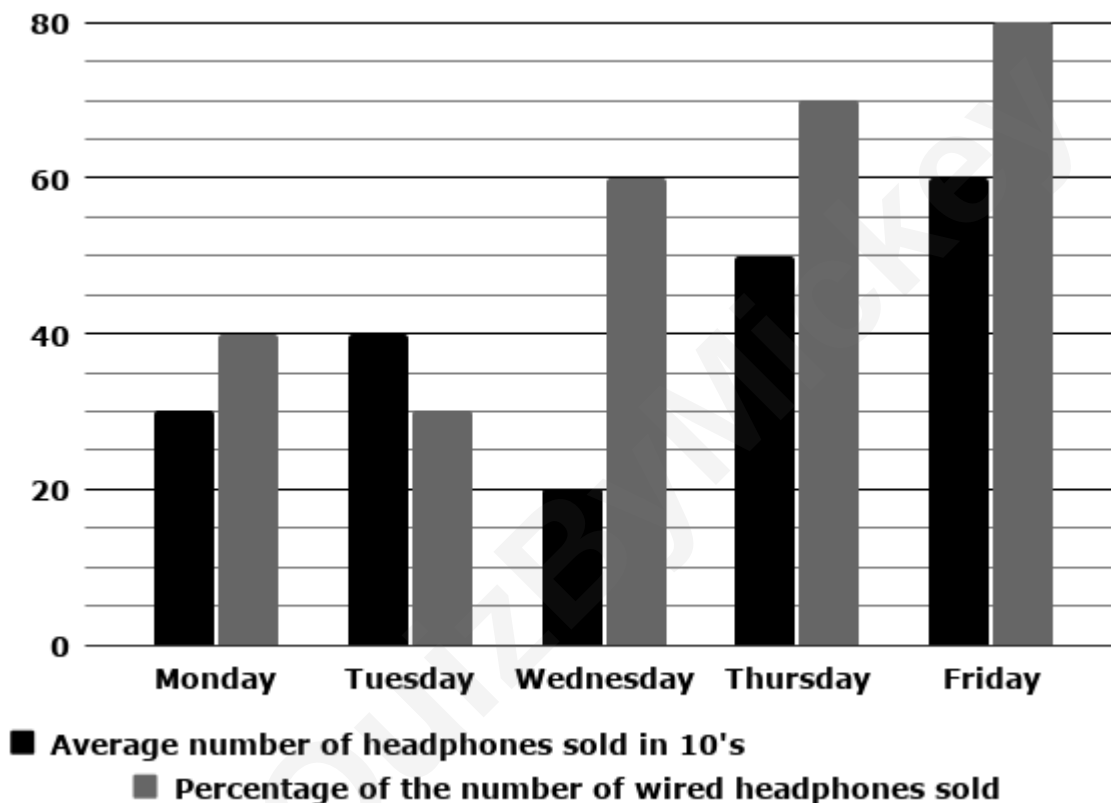


1. Questions

Study the following information carefully and answer the following questions.

The given bar graph shows the average number of headphones sold (in the 10's) on five different days namely Monday, Tuesday, Wednesday, Thursday and Friday respectively and also the percentage of the number of wired headphones sold on these five days.

Note: The total number of headphones sold on each day = The number of wired headphones sold + The number of wireless headphones sold.



The total number of headphones sold on Saturday is twice the difference between the total number of headphones sold on Tuesday and Wednesday. If 40% of the headphones sold on Saturday are wired, then find the number of wireless headphones sold on Saturday.

- a. 320
- b. 480
- c. 450
- d. 400
- e. 200

2. Questions

On Thursday, out of the total number of wired headphones manufactured, 30% were unsold and the rest of them were sold. If the number of wireless headphones unsold is 20 more than that of

wired headphones, then find the total number of wireless headphones manufactured on Thursday.

- a. 550
- b. 620
- c. 480
- d. 720
- e. 700

3. Questions

If the total number of headphones sold on Friday is increased by 25%, then find the total number of headphones sold on Friday after the increase.

- a. 1500
- b. 2100
- c. 1800
- d. 1440
- e. 1600

4. Questions

Find the difference between the sum of the number of wired headphones sold on Thursday and Friday together and the number of wireless headphones sold on Monday.

- a. 1250
- b. 1500
- c. 1300
- d. 1200
- e. 900

5. Questions

Find the ratio between the number of wired headphones sold on Wednesday to the number of wireless headphones sold on Tuesday.

- a. 7:3
- b. 3:1
- c. 3:7
- d. 2:1
- e. 4:3

6. Questions

Study the following information carefully and answer the following questions.

The given table chart shows the number of tigers in five different zoos namely A, B, C, D and E respectively and also the percentage of the number of lions in these five different zoos.

Note: The total number of animals in each zoo = The number of tigers + The number of lions.

Zoos	The number of tigers	Percentage of the number of lions
A	450	70%
B	600	40%
C	315	55%
D	800	20%
E	330	45%

The ratio of the number of lions in zoo A to F is 5:3 and the number of tigers in zoo F is $\frac{1}{7}$ less than that of lions. Find the total number of animals in zoo F.

- a. 650
- b. 1720
- c. 1170
- d. 990
- e. 780

7. Questions

If the average number of lions and elephants in zoo C is 500 and the ratio of the number of male to female elephants is 2:1. Find the number of male elephants.

- a. 180
- b. 210
- c. 400
- d. 80
- e. 410

8. Questions

Find the average number of animals in zoo B, C and D together.

- a. 800
- b. 720
- c. 1300
- d. 900
- e. 980

9. Questions

The number of lions in zoo B is what percentage of the total number of animals in zoo D?

- a. 60%
- b. 55%
- c. 45%
- d. 40%
- e. 25%

10. Questions

Find the ratio between the average number of tigers in zoo B and D to the $\frac{1}{9}$ th of the number of lions in zoo E.

- a. 77: 4
- b. 70: 3
- c. 34: 31
- d. 19: 7
- e. 3: 4

11. Questions

Following questions have two quantities as Quantity I and Quantity II. You have to determine the relationship between them and give answer as,

Quantity I: The income of A is 40% more than the income of B. The expenditure of A and B are 60% and 50% respectively. Then what percentage of savings of B is savings of A?

Quantity II: 55%

- a. Quantity I > Quantity II
- b. Quantity I $\geq$ Quantity II
- c. Quantity II > Quantity I
- d. Quantity II $\geq$ Quantity I
- e. Quantity I = Quantity II or Relation cannot be established

12. Questions

Quantity I: Rs. 2000 when invested at $r\%$ per annum compound interest, compounded annually for 2 years and the interest received is Rs. 880. Find the value of r .

Quantity II: A certain sum amounts to 5 times itself in 8 years when invested at $r\%$ per annum simple interest. Find the value of r .

- a. Quantity I > Quantity II
- b. Quantity I $\geq$ Quantity II
- c. Quantity II > Quantity I
- d. Quantity II $\geq$ Quantity I
- e. Quantity I = Quantity II or Relation cannot be established

13. Questions

Quantity I: The speed of a man increases by 3 km/hr after every 20 minutes. If the distance travelled by a man in 1st 20 minutes is 4 km, then find the total distance travelled by a man in 60 minutes.

Quantity II: 21 km

- a. Quantity I > Quantity II
- b. Quantity I $\geq$ Quantity II
- c. Quantity II > Quantity I
- d. Quantity II $\geq$ Quantity I
- e. Quantity I = Quantity II or Relation cannot be established

14. Questions

Vessel A contains milk and water, where milk is 12 litres more than water and the quantity of milk in vessel B is 25% more than that of vessel A. If the difference between the quantity of milk in vessel A and B is 28 litres, then find the quantity of water in vessel A (Vessel B contains only milk and water).

- a. 120 litres
- b. 100 litres
- c. 80 litres
- d. 60 litres
- e. 150 litres

15. Questions

Vicky bought two articles A and B at the same price. He marked article A at 25% and article B at 30% above its cost price and article A and B sold for a discount of Rs. 200 and Rs. 300 respectively.

If the selling price of both the articles is same, then find the cost price of article A.

- a. Rs. 4500
- b. Rs. 2000
- c. Rs. 3200
- d. Rs. 3000
- e. Rs. 5000

16. Questions

P and Q entered into a business by investing Rs. $(x + 800)$ and Rs. $6x$ respectively for 6 months and 15 months respectively. If the ratio of the profit share of P to Q is 3:25, then find the value of x .

- a. 2000
- b. 1800
- c. 2100
- d. 1000
- e. 800

17. Questions

Rs. 2640 were given among 3 persons namely P, Q and R. The amount received by P is 25% more than that of Q and 40% more than that of R. Find the amount received by P.

- a. Rs. 1150
- b. Rs. 950
- c. Rs. 1280
- d. Rs. 1050
- e. Rs. 750

18. Questions

The sum of the speed of the boat that covers 60 km upstream in 12 hours and 2D km downstream in 4 hours is $(D - 10)$ km/hr. Find the value of D.

- a. 25
- b. 10
- c. 30
- d. 40
- e. 50

19. Questions

Pipe A alone can fill a tank in 36 hours, while Pipe B alone can fill the same tank in 45 hours. The ratio of the efficiencies of Pipes B and C is 4:1. If Pipe A fills 25% of the tank, Pipes A and C together fill the next 30% of the tank, and Pipes A and B together fill the remaining portion of the tank, find the total time taken to fill the tank completely.

- a. 30 hours
- b. 27 hours
- c. 45 hours
- d. 25 hours
- e. 35 hours

20. Questions

There are four consecutive even numbers. The second even number is equal to 6 more than the square of 6. Find the difference between the 25% of the first and third even number together and the fourth even number.

- a. 30
- b. 27
- c. 35
- d. 25
- e. 20

21. Questions

What value should come in the place of (?) in the following questions.

$$14^2 + (24 * 18) - 323 = ? * 7 + 18$$

- a. 55
- b. 41
- c. 36
- d. 25
- e. 40

22. Questions

$$30\% \text{ of } 450 + 11 * 22 - ? = 15 * 12$$

- a. 210
- b. 188

- c. 201
- d. 197
- e. 230

23. Questions

$$(? + 15) * 2 + 90 * 2 = 320 + 60 * 2$$

- a. 115
- b. 90
- c. 125
- d. 156
- e. 88

24. Questions

$$(89 + 71 - 45) * 3 + ? = 720$$

- a. 568
- b. 375
- c. 210
- d. 670
- e. 420

25. Questions

$$?^2 - 14 = 19^2 - 15\% \text{ of } 400 + 17 * 5$$

- a. 18
- b. 25
- c. 15
- d. 10
- e. 20

26. Questions

What approximate value should come in the place of (?) in the following questions.

$$31.21 * 9.98 - \sqrt{196.02} * 5.11 = ? - 42.12$$

- a. 210
- b. 282

- c. 455
- d. 100
- e. 189

27. Questions

$$(19.99)^2 + \sqrt{1089.01} - 45.12\% \text{ of } 300.12 = ?$$

- a. 215
- b. 323
- c. 425
- d. 298
- e. 300

28. Questions

$$32.11 * 2.90 - 6.01^2 * 3.11 + ? = 70.33\% \text{ of } 300.01$$

- a. 222
- b. 245
- c. 188
- d. 190
- e. 255

29. Questions

$$22.21\% \text{ of } 500.13 + 80.12\% \text{ of } 899.89 - 9.11^2 = ?$$

- a. 817
- b. 749
- c. 819
- d. 245
- e. 200

30. Questions

$$(1728)^{1/3} - (256.12)^{1/2} * 2.01 = ?^2 - 119.89$$

- a. 10
- b. 5

- c. 6
- d. 8
- e. 21

31. Questions

What value should come in the place of (?) in the following number series?

900, 7200, ?, 4800, 2400

- a. 800
- b. 1550
- c. 1200
- d. 2000
- e. 1800

32. Questions

8, ?, 48, 192, 960, 5760

- a. 16
- b. 24
- c. 19
- d. 20
- e. 32

33. Questions

3, 4, ?, 28, 113, 566

- a. 9
- b. 16
- c. 22
- d. 10
- e. 12

34. Questions

19, 28, 39, ?, 67, 84

- a. 42
- b. 50

- c. 65
- d. 52
- e. 41

35. Questions

92, 193, 138, 239, ?, 285

- a. 184
- b. 205
- c. 215
- d. 172
- e. 164

36. Questions

Find out the wrong number in the following number series.

29, 54, 158, 379, 779, 1404

- a. 379
- b. 779
- c. 1404
- d. 29
- e. 158

37. Questions

727, 510, 341, 214, 123, 66

- a. 341
- b. 66
- c. 123
- d. 510
- e. 214

38. Questions

24, 60, 180, 634, 2520, 11340

- a. 180
- b. 24

- c. 2520
- d. 11340
- e. 634

39. Questions

34, 43, 55, 97, 265

- a. 43
- b. 55
- c. 97
- d. 265
- e. 34

40. Questions

527, 504, 473, 432, 381, 320

- a. 381
- b. 527
- c. 432
- d. 320
- e. 504

41. Questions

The following question contains two equations as I and II. You have to solve both equations and determine the relationship between them and give answer as,

i). $3x^2 + 15x + 18 = 0$

ii). $y^2 + 8y + 16 = 0$

- a. $x > y$
- b. $x \geq y$
- c. $x = y$ or relationship cannot be established.
- d. $x < y$
- e. $x \leq y$

42. Questions

i). $x^2 + 19x + 78 = 0$

ii). $y^2 + 13y + 42 = 0$

- a. $x > y$
- b. $x \geq y$
- c. $x = y$ or relationship cannot be established.
- d. $x < y$
- e. $x \leq y$

43. Questions

i). $x^2 - 43x + 462 = 0$

ii). $y^2 - 47y + 552 = 0$

- a. $x > y$
- b. $x \geq y$
- c. $x = y$ or relationship cannot be established.
- d. $x < y$
- e. $x \leq y$

44. Questions

i). $x^2 - 32x + 255 = 0$

ii). $y^2 - 29y + 210 = 0$

- a. $x > y$
- b. $x \geq y$
- c. $x = y$ or relationship cannot be established.
- d. $x < y$
- e. $x \leq y$

45. Questions

i). $x^2 - 3x - 270 = 0$

ii). $y^2 - 20y + 96 = 0$

- a. $x > y$
- b. $x \geq y$
- c. $x = y$ or relationship cannot be established.

d. $x < y$

e. $x \leq y$

Explanations:

1. Questions

On Monday:

The total number of headphones sold = $300 * 2 = 600$

The number of wired headphones sold = $600 * 40/100 = 240$

The number of wireless headphones sold = $600 * 60/100 = 360$

On Tuesday:

The total number of headphones sold = $400 * 2 = 800$

The number of wired headphones sold = $800 * 30/100 = 240$

The number of wireless headphones sold = $800 * 70/100 = 560$

On Wednesday:

The total number of headphones sold = $200 * 2 = 400$

The number of wired headphones sold = $400 * 60/100 = 240$

The number of wireless headphones sold = $400 * 40/100 = 160$

On Thursday:

The total number of headphones sold = $500 * 2 = 1000$

The number of wired headphones sold = $1000 * 70/100 = 700$

The number of wireless headphones sold = $1000 * 30/100 = 300$

On Friday:

The total number of headphones sold = $600 * 2 = 1200$

The number of wired headphones sold = $1200 * 80/100 = 960$

The number of wireless headphones sold = $1200 * 20/100 = 240$

Days	The total number of headphones sold	The number of wired headphones sold	The number of wireless headphones sold
Monday	600	240	360
Tuesday	800	240	560
Wednesday	400	240	160
Thursday	1000	700	300
Friday	1200	960	240

Answer: B

The total number of headphones sold on Saturday = $2 * (800 - 400) = 2 * 400 = 800$

The number of wireless headphones sold on Saturday = $800 * 60/100 = 480$

Hence, the correct answer is option B.

2. Questions

On Monday:

The total number of headphones sold = $300 * 2 = 600$

The number of wired headphones sold = $600 * 40/100 = 240$

The number of wireless headphones sold = $600 * 60/100 = 360$

On Tuesday:

The total number of headphones sold = $400 * 2 = 800$

The number of wired headphones sold = $800 * 30/100 = 240$

The number of wireless headphones sold = $800 * 70/100 = 560$

On Wednesday:

The total number of headphones sold = $200 * 2 = 400$

The number of wired headphones sold = $400 * 60/100 = 240$

The number of wireless headphones sold = $400 * 40/100 = 160$

On Thursday:

The total number of headphones sold = $500 * 2 = 1000$

The number of wired headphones sold = $1000 * 70/100 = 700$

The number of wireless headphones sold = $1000 * 30/100 = 300$

On Friday:

The total number of headphones sold = $600 * 2 = 1200$

The number of wired headphones sold = $1200 * 80/100 = 960$

The number of wireless headphones sold = $1200 * 20/100 = 240$

Days	The total number of headphones sold	The number of wired headphones sold	The number of wireless headphones sold
Monday	600	240	360
Tuesday	800	240	560
Wednesday	400	240	160
Thursday	1000	700	300
Friday	1200	960	240

Answer: B

The total number of wired headphones manufactured on Thursday = $700 * 100/70 = 1000$

The number of wired headphones unsold on Thursday = $1000 - 700 = 300$

The number of wireless headphones unsold on Thursday = $(300 + 20) = 320$

The total number of wireless headphones manufactured on Thursday = $(320 + 300) = 620$

Hence, the correct answer is option B.

3. Questions

On Monday:

The total number of headphones sold = $300 * 2 = 600$

The number of wired headphones sold = $600 * 40/100 = 240$

The number of wireless headphones sold = $600 * 60/100 = 360$

On Tuesday:

The total number of headphones sold = $400 * 2 = 800$

The number of wired headphones sold = $800 * 30/100 = 240$

The number of wireless headphones sold = $800 * 70/100 = 560$

On Wednesday:

The total number of headphones sold = $200 * 2 = 400$

The number of wired headphones sold = $400 * 60/100 = 240$

The number of wireless headphones sold = $400 * 40/100 = 160$

On Thursday:

The total number of headphones sold = $500 * 2 = 1000$

The number of wired headphones sold = $1000 * 70/100 = 700$

The number of wireless headphones sold = $1000 * 30/100 = 300$

On Friday:

The total number of headphones sold = $600 * 2 = 1200$

The number of wired headphones sold = $1200 * 80/100 = 960$

The number of wireless headphones sold = $1200 * 20/100 = 240$

Days	The total number of headphones sold	The number of wired headphones sold	The number of wireless headphones sold
Monday	600	240	360
Tuesday	800	240	560
Wednesday	400	240	160
Thursday	1000	700	300
Friday	1200	960	240

Answer: A

The total number of headphones sold on Friday after the increase = $1200 * 125/100 = 1500$

Hence, the correct answer is option A.

4. Questions

On Monday:

The total number of headphones sold = $300 * 2 = 600$

The number of wired headphones sold = $600 * 40/100 = 240$

The number of wireless headphones sold = $600 * 60/100 = 360$

On Tuesday:

The total number of headphones sold = $400 * 2 = 800$

The number of wired headphones sold = $800 * 30/100 = 240$

The number of wireless headphones sold = $800 * 70/100 = 560$

On Wednesday:

The total number of headphones sold = $200 * 2 = 400$

The number of wired headphones sold = $400 * 60/100 = 240$

The number of wireless headphones sold = $400 * 40/100 = 160$

On Thursday:

The total number of headphones sold = $500 * 2 = 1000$

The number of wired headphones sold = $1000 * 70/100 = 700$

The number of wireless headphones sold = $1000 * 30/100 = 300$

On Friday:

The total number of headphones sold = $600 * 2 = 1200$

The number of wired headphones sold = $1200 * 80/100 = 960$

The number of wireless headphones sold = $1200 * 20/100 = 240$

Days	The total number of headphones sold	The number of wired headphones sold	The number of wireless headphones sold
Monday	600	240	360
Tuesday	800	240	560
Wednesday	400	240	160
Thursday	1000	700	300
Friday	1200	960	240

Answer: C

The sum of the number of wired headphones sold on Thursday and Friday = $(700 + 960) = 1660$ 

The number of wireless headphones sold on Monday = 360

Required difference = $(1660 - 360) = 1300$

Hence, the correct answer is option C.

5. Questions

On Monday:

The total number of headphones sold = $300 * 2 = 600$

The number of wired headphones sold = $600 * 40/100 = 240$

The number of wireless headphones sold = $600 * 60/100 = 360$

On Tuesday:

The total number of headphones sold = $400 * 2 = 800$

The number of wired headphones sold = $800 * 30/100 = 240$

The number of wireless headphones sold = $800 * 70/100 = 560$

On Wednesday:

The total number of headphones sold = $200 * 2 = 400$

The number of wired headphones sold = $400 * 60/100 = 240$

The number of wireless headphones sold = $400 * 40/100 = 160$

On Thursday:

The total number of headphones sold = $500 * 2 = 1000$

The number of wired headphones sold = $1000 * 70/100 = 700$

The number of wireless headphones sold = $1000 * 30/100 = 300$

On Friday:

The total number of headphones sold = $600 * 2 = 1200$

The number of wired headphones sold = $1200 * 80/100 = 960$

The number of wireless headphones sold = $1200 * 20/100 = 240$

Days	The total number of headphones sold	The number of wired headphones sold	The number of wireless headphones sold
Monday	600	240	360
Tuesday	800	240	560
Wednesday	400	240	160
Thursday	1000	700	300
Friday	1200	960	240

Answer: C

The number of wired headphones sold on Wednesday = 240

The number of wireless headphones sold on Tuesday = 560

Required ratio = $240 : 560 = 12 : 28 = 3 : 7$

Hence, the correct answer is option C.

6. Questions

In zoo A:

The number of tigers = 450

The total number of animals = $450 \times 100/30 = 1500$

The number of lions = $(1500 - 450) = 1050$

In zoo B:

The number of tigers = 600

The total number of animals = $600 \times 10/60 = 1000$

The number of lions = $(1000 - 600) = 400$

In zoo C:

The number of tigers = 315

The total number of animals = $315 \times 100/45 = 700$

The number of lions = $(700 - 315) = 385$

In zoo D:

The number of tigers = 800

The total number of animals = $800 * 100/80 = 1000$

The number of lions = $(1000 - 800) = 200$

In zoo E:

The number of tigers = 330

The total number of animals = $330 * 100/55 = 600$

The number of lions = $600 - 330 = 270$

zoos	The number of tigers	The number of lions	The total number of animals
A	450	1050	1500
B	600	400	1000
C	315	385	700
D	800	200	1000
E	330	270	600

Answer: C

The number of lions in zoo A = 1050

The number of lions in zoo F = $1050 * 3/5 = 210 * 3 = 630$

The number of tigers in zoo F = $630 - 630/7 = (630 - 90) = 540$

Required sum = $(540 + 630) = 1170$

Hence, the correct answer is option C.

7. Questions

In zoo A:

The number of tigers = 450

The total number of animals = $450 * 100/30 = 1500$

The number of lions = $(1500 - 450) = 1050$

In zoo B:

The number of tigers = 600

The total number of animals = $600 * 10/60 = 1000$

The number of lions = $(1000 - 600) = 400$

In zoo C:

The number of tigers = 315

The total number of animals = $315 * 100/45 = 700$

The number of lions = $(700 - 315) = 385$

In zoo D:

The number of tigers = 800

The total number of animals = $800 * 100/80 = 1000$

The number of lions = $(1000 - 800) = 200$

In zoo E:

The number of tigers = 330

The total number of animals = $330 * 100/55 = 600$

The number of lions = $600 - 330 = 270$

zoos	The number of tigers	The number of lions	The total number of animals
A	450	1050	1500
B	600	400	1000
C	315	385	700
D	800	200	1000
E	330	270	600

Answer: E

The sum of the number of lions and elephants in zoo C = $500 * 2 = 1000$

The number of elephants in zoo C = $1000 - 385 = 615$

The number of male elephants in zoo C = $615 * 2/3 = 410$

Hence, the correct answer is option E.

8. Questions

In zoo A:

The number of tigers = 450

The total number of animals = $450 * 100/30 = 1500$

The number of lions = $(1500 - 450) = 1050$

In zoo B:

The number of tigers = 600

The total number of animals = $600 * 10/60 = 1000$

The number of lions = $(1000 - 600) = 400$

In zoo C:

The number of tigers = 315

The total number of animals = $315 * 100/45 = 700$

The number of lions = $(700 - 315) = 385$

In zoo D:

The number of tigers = 800

The total number of animals = $800 * 100/80 = 1000$

The number of lions = $(1000 - 800) = 200$

In zoo E:

The number of tigers = 330

The total number of animals = $330 * 100/55 = 600$

The number of lions = $600 - 330 = 270$

zoos	The number of tigers	The number of lions	The total number of animals
A	450	1050	1500
B	600	400	1000
C	315	385	700
D	800	200	1000
E	330	270	600

Answer: D

The sum of the total number of animals in zoo B, C and D together = $(1000 + 700 + 1000) = 2700$

Required average = $2700/3 = 900$

Hence, the correct answer is option D.

9. Questions

In zoo A:

The number of tigers = 450

The total number of animals = $450 * 100/30 = 1500$

The number of lions = $(1500 - 450) = 1050$

In zoo B:

The number of tigers = 600

The total number of animals = $600 * 10/60 = 1000$

The number of lions = $(1000 - 600) = 400$

In zoo C:

The number of tigers = 315

The total number of animals = $315 * 100/45 = 700$

The number of lions = $(700 - 315) = 385$

In zoo D:

The number of tigers = 800

The total number of animals = $800 * 100/80 = 1000$

The number of lions = $(1000 - 800) = 200$

In zoo E:

The number of tigers = 330

The total number of animals = $330 * 100/55 = 600$

The number of lions = $600 - 330 = 270$

zoos	The number of tigers	The number of lions	The total number of animals
A	450	1050	1500
B	600	400	1000
C	315	385	700
D	800	200	1000
E	330	270	600

Answer: D

The number of lions in zoo B = 400

The total number of animals in zoo D = 1000

Required percentage = $(400/1000) * 100 = 40\%$

Hence, the correct answer is option D.

10. Questions

In zoo A:

The number of tigers = 450

The total number of animals = $450 * 100/30 = 1500$

The number of lions = $(1500 - 450) = 1050$

In zoo B:

The number of tigers = 600

The total number of animals = $600 * 10/60 = 1000$

The number of lions = $(1000 - 600) = 400$

In zoo C:

The number of tigers = 315

The total number of animals = $315 * 100/45 = 700$

The number of lions = $(700 - 315) = 385$

In zoo D:

The number of tigers = 800

The total number of animals = $800 * 100/80 = 1000$

The number of lions = $(1000 - 800) = 200$

In zoo E:

The number of tigers = 330

The total number of animals = $330 * 100/55 = 600$

The number of lions = $600 - 330 = 270$

zoos	The number of tigers	The number of lions	The total number of animals
A	450	1050	1500
B	600	400	1000
C	315	385	700
D	800	200	1000
E	330	270	600

Answer: B

The average number of tigers in zoo B and D = $(600 + 800)/2 = 700$

1/9th of the number of lions in zoo E = $270/9 = 30$

Required ratio = $700:30 = 70:3$

Hence, the correct answer is option B.

11. Questions**Answer: A****Quantity I:**

The ratio of the income of A to B = $140:100 = 7x:5x$

Let, the income of A and B be Rs. $7x$ and Rs. $5x$ respectively.

The expenditure of A = $7x * 60/100 = \text{Rs. } 4.2x$

The savings of A = $7x - 4.2x = \text{Rs. } 2.8x$

The expenditure of B = $5x * 50/100 = \text{Rs. } 2.5x$

The savings of B = $5x * 50/100 = \text{Rs. } 2.5x$

Required percentage = $(2.8x/2.5x) * 100 = 112\%$

Quantity I: 112%

Quantity II: 55%

Thus, Quantity I > Quantity II

Hence, the correct answer is option A.

12. Questions**Answer: C****Quantity II:**

$$CI = P * (1 + R/100)^n - P$$

$$880 = 2000 * (1 + r/100)^2 - 2000$$

$$2880/2000 = (1 + r/100)^2$$

$$1.44 = (1 + r/100)^2$$

$$r = 120 - 100$$

$$r = 20$$

Quantity I: 20

Quantity II:

$$SI = \text{PNR}/100$$

Let, the sum = Rs. x

$$4x = x * 8/100 * r$$

$$400/8 = r$$

$$r = 50$$

Quantity II: 50

Thus, Quantity I < Quantity II

Hence, the correct answer is option C.

13. Questions

Answer: C

Quantity I:

The speed of the man in 1st 20 minutes = $60 * 4/20 = 12$ km/hr

The distance covered by man in 1st 20 minutes = $12 * (20/60) = 4$ km

The distance covered by man in 2nd 20 minutes = $15 * 20/60 = 5$ km

The distance covered by man in 3rd 20 minutes = $18 * 20/60 = 6$ km

Required distance = $(4 + 5 + 6) = 15$ km

Quantity I: 15 km

Quantity II: 21 km

Thus, Quantity II > Quantity I

Hence, the correct answer is option C.

14. Questions

Answer: B

Let, the initial quantity of water in vessel A = x litres

The initial quantity of milk in vessel A = $(x + 12)$ litres

The initial quantity of milk in vessel B = $(x + 12) * 125/100 = (1.25x + 15)$ litres

$$(1.25x + 15) - x - 12 = 28$$

$$0.25x + 3 = 28$$

$$0.25x = 25$$

$$x = 100$$

The initial quantity of water in vessel A = 100 litres

Hence, the correct answer is option B.

15. Questions**Answer: B**

Let, the cost price of both the articles = Rs. $100x$

The marked price of the article A = $100x * 125/100 = \text{Rs. } 125x$

The marked price of the article B = $100x * 130/100 = \text{Rs. } 130x$

$$(125x - 200) = (130x - 300)$$

$$125x - 130x = -300 + 200$$

$$5x = 100$$

$$x = 20$$

The cost price of article A = $100 * 20 = \text{Rs. } 2000$

Hence, the correct answer is option B.

16. Questions**Answer: D**

The ratio of the profit share of P to Q = $(x + 800) * 6 : (6x * 15)$

$$= (x + 800) : 15x$$

$$(x + 800)/15x = 3/25$$

$$(x + 800)/3x = 3/5$$

$$5x + 4000 = 9x$$

$$4x = 4000$$

$$x = 1000$$

Hence, the correct answer is option D.

17. Questions**Answer: D**

The ratio of the amount received by P to Q = $125 : 100 = 5 : 4$

The ratio of the amount received by P to R = $140 : 100 = 7 : 5$

The ratio of the amount received by P, Q to R = $35 : 28 : 25$

The amount received by P = $2640 * 35/88 = \text{Rs. } 1050$

Hence, the correct answer is option D.

18. Questions**Answer: C**

Speed = Distance/Time

$$(60/12) + (2D/4) = (D - 10)$$

$$5 + D/2 = D - 10$$

$$10 + D = 2D - 20$$

$$D = 30$$

Hence, the correct answer is option C.

19. Questions

Answer: B

Let, the total work = 180 units (LCM of 36 and 45)

The efficiency of pipe A = $180/36 = 5$ units/hour

The efficiency of pipe B = $180/45 = 4$ units/hour

The efficiency of pipe C = $4 * 1/4 = 1$ units/hour

The 25% of the work = $180 * 25/100 = 45$ units

The 30% of the work = $180 * 30/100 = 54$ units

The remaining work = $(180 - 45 - 54) = 81$ units

Required time taken = $(45/5) + (54/6) + (81/9) = (9 + 9 + 9) = 27$ hours

Hence, the correct answer is option B.

20. Questions

Answer: D

Let, the four consecutive even numbers be x , $(x + 2)$, $(x + 4)$ and $(x + 6)$ respectively.

$$x + 2 = 6^2 + 6$$

$$x + 2 = 36 + 6$$

$$x + 2 = 42$$

$$x = 40$$

The four even numbers = 40, 42, 44, 46

Required difference = $46 - (40 + 44) * 25/100$

$$= 46 - 84 * 1/4$$

$$= 46 - 21$$

$$= 25$$

Hence, the correct answer is option D.

21. Questions

Answer: B

$$14^2 + (24 * 18) - 323 = ? * 7 + 18$$

$$196 + 432 - 323 - 18 = ? * 7$$

$$287/7 = ?$$

$$? = 41$$

Hence, the correct answer is option B.

22. Questions

Answer: D

$$30\% \text{ of } 450 + 11 * 22 - ? = 15 * 12$$

$$135 + 242 - ? = 180$$

$$377 - ? = 180$$

$$? = 197$$

Hence, the correct answer is option D.

23. Questions

Answer: A

$$(? + 15) * 2 + 90 * 2 = 320 + 60 * 2$$

$$(? + 15) * 2 + 180 = 320 + 120$$

$$(? + 15) * 2 + 180 = 440$$

$$(? + 15) * 2 = 260$$

$$? + 15 = 130$$

$$? = 115$$

Hence, the correct answer is option A.

24. Questions

Answer: B

$$(89 + 71 - 45) * 3 + ? = 720$$

$$115 * 3 + ? = 720$$

$$345 + ? = 720$$

$$? = 375$$

Hence, the correct answer is option B.

25. Questions

Answer: E

$$?^2 - 14 = 19^2 - 15\% \text{ of } 400 + 17 * 5$$

$$?^2 - 14 = 361 - 60 + 85$$

$$?^2 = 386 + 14$$

$$?^2 = 400$$

$$? = 20$$

Hence, the correct answer is option E.

26. Questions

Answer: B

$$31.21 * 9.98 - \sqrt{196.02} * 5.11 = ? - 42.12$$

$$31 * 10 - \sqrt{196} * 5 + 42 = ?$$

$$310 - 14 * 5 + 42 = ?$$

$$310 - 70 + 42 = ?$$

$$? = 282$$

Hence, the correct answer is option B.

27. Questions

Answer: D

$$(19.99)^2 + \sqrt{1089.01} - 45.12\% \text{ of } 300.12 = ?$$

$$20^2 + 33 - 45\% \text{ of } 300 = ?$$

$$400 + 33 - 135 = ?$$

$$? = 298$$

Hence, the correct answer is option D.

28. Questions

Answer: A

$$32.11 * 2.90 - 6.01^2 * 3.11 + ? = 70.33\% \text{ of } 300.01$$

$$32 * 3 - 6^2 * 3 + ? = 70\% \text{ of } 300$$

$$96 - 36 * 3 + ? = 210$$

$$96 - 108 + ? = 210$$

$$-12 + ? = 210$$

$$? = 222$$

Hence, the correct answer is option A.

29. Questions**Answer: B**

$$22.21\% \text{ of } 500.13 + 80.12\% \text{ of } 899.89 - 9.11^2 = ?$$

$$22\% \text{ of } 500 + 80\% \text{ of } 900 - 9^2 = ?$$

$$110 + 720 - 81 = ?$$

$$? = 749$$

Hence, the correct answer is option B.

30. Questions**Answer: A**

$$(1728)^{1/3} - (256.12)^{1/2} * 2.01 = ?^2 - 119.89$$

$$1728^{1/3} - 256^{1/2} * 2 = ?^2 - 120$$

$$12 - 16 * 2 = ?^2 - 120$$

$$12 - 32 = ?^2 - 120$$

$$-20 + 120 = ?^2$$

$$?^2 = 100$$

$$? = 10$$

Hence, the correct answer is option A.

31. Questions**Answer: C**

$$900 * 8 = 7200$$

$$7200/6 = \mathbf{1200}$$

$$1200 * 4 = 4800$$

$$4800/2 = 2400$$

Hence, the correct answer is option C.

32. Questions**Answer: A**

$$8 * 2 = \mathbf{16}$$

$$16 * 3 = 48$$

$$48 * 4 = 192$$

$$192 * 5 = 960$$

$$960 * 6 = 5760$$

Hence, the correct answer is option A.

33. Questions

Answer: A

$$3 * 1 + 1 = 4$$

$$4 * 2 + 1 = \mathbf{9}$$

$$9 * 3 + 1 = 28$$

$$28 * 4 + 1 = 113$$

$$113 * 5 + 1 = 566$$

Hence, the correct answer is option A.

34. Questions

Answer: D

$$4^2 + 3 = 19$$

$$5^2 + 3 = 28$$

$$6^2 + 3 = 39$$

$$7^2 + 3 = \mathbf{52}$$

$$8^2 + 3 = 67$$

$$9^2 + 3 = 84$$

OR

$$19 \quad 28 \quad 39 \quad \mathbf{52} \quad 67 \quad 84$$

$$9 \quad 11 \quad 13 \quad 15 \quad 17$$

Hence, the correct answer is option D.

35. Questions

Answer: A

$$92 + 101 = 193$$

$$193 - 55 = 138$$

$$138 + 101 = 239$$

$$239 - 55 = \mathbf{184}$$

$$184 + 101 = 285$$

Hence, the correct answer is option A.

36. Questions

Answer: E

29	54	154	379	779	1404
+ 25	+100	+ 225	+400	+625	
5^2	10^2	15^2	20^2	25^2	

Hence, the correct answer is option E.

37. Questions

Answer: B

$$9^3 - 2 = 727$$

$$8^3 - 2 = 510$$

$$7^3 - 2 = 341$$

$$6^3 - 2 = 214$$

$$5^3 - 2 = 123$$

$$4^3 - 2 = \mathbf{62}$$

Hence, the correct answer is option B.

38. Questions

Answer: E

$$24 * 2.5 = 60$$

$$60 * 3 = 180$$

$$180 * 3.5 = \mathbf{630}$$

$$630 * 4 = 2520$$

$$2520 * 4.5 = 11340$$

Hence, the correct answer is option E.

39. Questions

Answer: A

34	41	55	97	265
+ 7	+14	+ 42	+168	
*2	*3	*4		

Hence, the correct answer is option A.

40. Questions

Answer: B

$$525 - 21 = 504$$

$$504 - 31 = 473$$

$$473 - 41 = 432$$

$$432 - 51 = 381$$

$$381 - 61 = 320$$

Hence, the correct answer is option B.

41. Questions

Answer: A

$$3x^2 + 15x + 18 = 0$$

$$3x^2 + 6x + 9x + 18 = 0$$

$$3x(x+2) + 9(x+2) = 0$$

$$(3x + 9)(x + 2) = 0$$

$$x = -3, -2$$

$$y^2 + 8y + 16 = 0$$

$$y^2 + 4y + 4y + 16 = 0$$

$$y(y + 4) + 4(y + 4) = 0$$

$$(y + 4)(y + 4) = 0$$

$$y = -4, -4$$

Thus, $x > y$

Hence, the correct answer is option A.

42. Questions

Answer: C

$$x^2 + 19x + 78 = 0$$

$$x^2 + 13x + 6x + 78 = 0$$

$$x(x + 13) + 6(x + 13) = 0$$

$$(x + 13)(x + 6) = 0$$

$$x = -13, -6$$

$$y^2 + 13y + 42 = 0$$

$$y^2 + 6y + 7y + 42 = 0$$

$$y(y + 6) + 7(y + 6) = 0$$

$$(y + 6)(y + 7) = 0$$

$$y = -6, -7$$

Thus, the relationship between x and y cannot be established.

Hence, the correct answer is option C.

43. Questions

Answer: D

$$x^2 - 43x + 462 = 0$$

$$x^2 - 21x - 22x + 462 = 0$$

$$x(x - 21) - 22(x - 21) = 0$$

$$(x - 21)(x - 22) = 0$$

$$x = 21, 22$$

$$y^2 - 47y + 552 = 0$$

$$y^2 - 23y - 24y + 552 = 0$$

$$y(y - 23) - 24(y - 23) = 0$$

$$(y - 23)(y - 24) = 0$$

$$y = 23, 24$$

Thus, $x < y$

Hence, the correct answer is option D.

44. Questions

Answer: B

$$x^2 - 32x + 255 = 0$$

$$x^2 - 17x - 15x + 255 = 0$$

$$x(x - 17) - 15(x - 17) = 0$$

$$(x - 17)(x - 15) = 0$$

$$x = 17, 15$$

$$y^2 - 29y + 210 = 0$$

$$y^2 - 14y - 15y + 210 = 0$$

$$y(y - 14) - 15(y - 14) = 0$$

$$(y - 14)(y - 15) = 0$$

$$y = 14, 15$$

Thus, $x \geq y$

Hence, the correct answer is option B.

45. Questions

Answer: C

$$x^2 - 3x - 270 = 0$$

$$x^2 - 18x + 15x - 270 = 0$$

$$x(x - 18) + 15(x - 18) = 0$$

$$(x - 18)(x + 15) = 0$$

$$x = 18, -15$$

$$y^2 - 20y + 96 = 0$$

$$y^2 - 12y - 8y + 96 = 0$$

$$y(y - 12) - 8(y - 12) = 0$$

$$(y - 12)(y - 8) = 0$$

$$y = 12, 8$$

Thus, the relationship between x and y cannot be established

Hence, the correct answer is option C.